MATH1101: Differential Calculus & Analytical Geometry

Complete Exam Notes · 1st Year 1st Semester · PUST · Module 8 (Units 22, 23 & 24): Sphere, Cone, Cylinder & Central Conicoids

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Module 8 Overview

Units Covered in Module 8:

- **Unit 22:** The Sphere (General Equation, 4-Point Sphere, Plane Section Circle, Tangent Plane, Orthogonal & Coaxial Spheres).
- **Unit 23:** The Cone, Right Circular Cone, and Right Circular Cylinder.
- **Unit 24:** General Second-Degree Equation in 3D, Reduction to Standard Forms, and Identification of Conicoids.

Master Roadmap (MATH 1101)

Mod.	Units & Core Syllabus Topics	Main Sources	Status
1	Units 1–3: Functions, Domain/Range, Graphs, Inverse, Limits & Continuity	Class Notes · Anton Ch. 0, 1 · PUST 20–23	Module 1
2	Units 4–5: Differentiability, Geometric Meaning, Extrema, y_n & Leibniz	Class Notes · Anton Ch. 2, 4 · PUST 20–23	Module 2
3	Units 6–9: Rolle's & MVT, Taylor/Maclaurin, L'Hôpital & Curve Tracing	Class Notes · Anton Ch. 3, 4, 9 · Handout	Module 3
4	Units 10–13: Tangents, Asymptotes, Curvature, Partial Deriv., Euler & Jacobian	Akib Notes · Anton Ch. 13 · PUST 20–23	Module 4
5	Units 14–15: 2D Coordinates, Transformation of Axes & Pair of Straight Lines	Anton Ch. 10 · Rahman & Bhattacharjee	Module 5
6	Units 16–18: Conic Reduction, Parabola, Ellipse, Hyperbola & System of Circles	Anton Ch. 10 · Rahman & Bhattacharjee	Module 6
7	Units 19–21: 3D Coordinates, DCs/DRs, Projections, Planes & Skew Lines (S.D.)	Anton Ch. 11 · Rahman & Bhattacharjee	Module 7
8	Units 22–24: Sphere, Cone, Right Circular Cylinder & 3D Central Conicoids	Anton Ch. 11 · Rahman & Bhattacharjee	✓ **This file**

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1 Unit 22: The Sphere and Systems of Spheres

1.1 Standard Equations and Properties

1. **General Equation:** $S \equiv x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$, with Center $C = (-u, -v, -w)$ and Radius $R = \sqrt{u^2 + v^2 + w^2 - d}$.
2. **Diameter Form:** $(x - x_1)(x - x_2) + (y - y_1)(y - y_2) + (z - z_1)(z - z_2) = 0$.
3. **Plane Section of a Sphere (Circle in 3D):** A plane $P = 0$ at distance $p < R$ from the sphere's center C cuts the sphere in a circle of radius $r = \sqrt{R^2 - p^2}$ whose center N is the foot of the perpendicular from C onto $P = 0$.
4. **Sphere Through a Given Circle ($S = 0, P = 0$):** $S + \lambda P = 0$.
5. **Tangent Plane at (x_1, y_1, z_1):** $xx_1 + yy_1 + zz_1 + u(x + x_1) + v(y + y_1) + w(z + z_1) + d = 0$.
6. **Orthogonal Spheres:** $2u_1u_2 + 2v_1v_2 + 2w_1w_2 = d_1 + d_2$.

PUST Past Exam Questions on the Sphere [5 Marks Each]

Problem A (Center & Radius of Circle of Intersection): Find the center and radius of the circle $x^2 + y^2 + z^2 - 2y - 4z - 11 = 0,\ x + 2y + 2z = 15$.

Solution (Exam Style):

Sphere center $C = (0, 1, 2)$, Sphere radius $R = \sqrt{0 + 1 + 4 + 11} = 4$

Distance from C to plane, $p = \dfrac{|0 + 2(1) + 2(2) - 15|}{\sqrt{1^2 + 2^2 + 2^2}} = \dfrac{9}{3} = 3$

$\implies$ Circle radius, $r = \sqrt{R^2 - p^2} = \sqrt{16 - 9} = \sqrt{7}$.

Normal line through $C(0, 1, 2)$ perpendicular to $x + 2y + 2z = 15$:

$$\frac{x - 0}{1} = \frac{y - 1}{2} = \frac{z - 2}{2} = t \implies N = (t,\ 2t + 1,\ 2t + 2)$$

Since N lies on $x + 2y + 2z = 15$: $t + 2(2t + 1) + 2(2t + 2) = 15$

$\implies 9t + 6 = 15 \implies t = 1 \implies$ Circle center $N = (1, 3, 4)$.

Problem B (Spheres Through a Circle Touching a Plane): Find the spheres through $x^2+y^2+z^2 = 5,\ x+2y+3z = 3$ that touch $4x + 3y = 15$.

Solution (Exam Style):

Sphere family $S + \lambda P = 0$: $x^2 + y^2 + z^2 + \lambda x + 2\lambda y + 3\lambda z - (5 + 3\lambda) = 0$

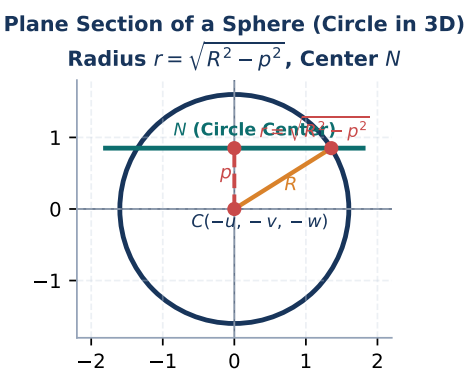
Center $C = \left(-\dfrac{\lambda}{2},\ -\lambda,\ -\dfrac{3\lambda}{2}\right)$, $R = \sqrt{\dfrac{7\lambda^2}{2} + 3\lambda + 5}$

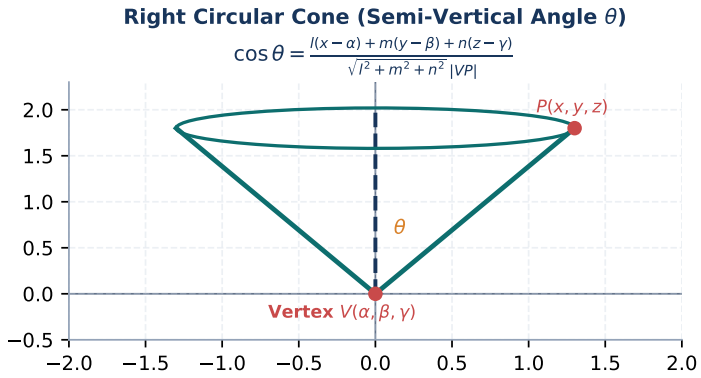
Tangency condition $p = R$: $\dfrac{|-5\lambda - 15|}{5} = \sqrt{\dfrac{7\lambda^2}{2} + 3\lambda + 5}$

$\implies (\lambda + 3)^2 = \dfrac{7\lambda^2}{2} + 3\lambda + 5 \implies 5\lambda^2 - 6\lambda - 8 = 0$

$\implies \lambda = 2,\ -\dfrac{4}{5}$

$\therefore x^2 + y^2 + z^2 + 2x + 4y + 6z - 11 = 0$ and $5(x^2 + y^2 + z^2) - 4x - 8y - 12z - 13 = 0$.





2 Unit 23: The Cone, Right Circular Cone, and Cylinder

2.1 Formulas for Cones and Cylinders

1. **Homogeneous Cone with Vertex at** $(0,0,0)$**:** $ax^2+by^2+cz^2+2fyz+2gzx+2hxy=0$. It has three mutually perpendicular generators if $a+b+c=0$.
2. **Right Circular Cone (Vertex** (α,β,γ)**, Axis DRs** (l,m,n)**, Semi-Vertical Angle** θ**):**
$$\big[l(x-\alpha)+m(y-\beta)+n(z-\gamma)\big]^2=(l^2+m^2+n^2)\big[(x-\alpha)^2+(y-\beta)^2+(z-\gamma)^2\big]\cos^2\theta.$$
3. **Right Circular Cylinder (Axis through** (α,β,γ) **with DRs** (l,m,n) **and Radius** R**):**
$$(x-\alpha)^2+(y-\beta)^2+(z-\gamma)^2-\frac{\big[l(x-\alpha)+m(y-\beta)+n(z-\gamma)\big]^2}{l^2+m^2+n^2}=R^2.$$

PUST Past Exam Questions on Cone and Cylinder [5 Marks Each]

Problem A (Cone with Given Vertex & Guiding Curve): Find the cone with vertex $(1,2,3)$ and guiding curve $x^2+y^2=4,\ z=0$.

Solution (Exam Style):

Generator through $(1,2,3)$: $\dfrac{x-1}{l}=\dfrac{y-2}{m}=\dfrac{z-3}{n}$

At $z=0$: $x=1-\dfrac{3l}{n}, \qquad y=2-\dfrac{3m}{n}$

Since it lies on $x^2+y^2=4$: $\left(1-\dfrac{3l}{n}\right)^2+\left(2-\dfrac{3m}{n}\right)^2=4$

$$\Longrightarrow \left(1-\frac{3(x-1)}{z-3}\right)^2+\left(2-\frac{3(y-2)}{z-3}\right)^2=4 \quad [\text{Using } \tfrac{l}{n}=\tfrac{x-1}{z-3},\ \tfrac{m}{n}=\tfrac{y-2}{z-3}]$$
$$\Longrightarrow (z-3x)^2+(2z-3y)^2=4(z-3)^2$$
$$\Longrightarrow 9x^2+9y^2+z^2-6xz-12yz+24z-36=0.$$

Problem B (Right Circular Cylinder): Find the right circular cylinder of radius $R=2$ with axis $\dfrac{x-1}{2}=\dfrac{y-2}{1}=\dfrac{z-3}{2}$.

Solution (Exam Style):

$$(x-1)^2+(y-2)^2+(z-3)^2-\left[\frac{2(x-1)+1(y-2)+2(z-3)}{3}\right]^2=2^2$$
$$\Longrightarrow 9\big[(x-1)^2+(y-2)^2+(z-3)^2\big]-(2x+y+2z-10)^2=36$$
$$\Longrightarrow 5x^2+8y^2+5z^2-4xy-4yz-8zx+22x-16y-14z+26=0.$$

3 Unit 24: Conicoids and Reduction of 3D Second-Degree Equations

Quadric Surface	Standard Equation	Key Geometric Features
1. Ellipsoid	$\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$	Bounded central conicoid; all plane sections are ellipses.
2. Hyperboloid of 1 Sheet	$\frac{x^2}{a^2}+\frac{y^2}{b^2}-\frac{z^2}{c^2}=1$	Connected central conicoid (1 minus sign); doubly ruled surface.
3. Hyperboloid of 2 Sheets	$\frac{x^2}{a^2}-\frac{y^2}{b^2}-\frac{z^2}{c^2}=1$	Two disconnected sheets (2 minus signs).
4. Elliptic Paraboloid	$\frac{x^2}{a^2}+\frac{y^2}{b^2}=2cz$	Non-central conicoid opening along the z-axis.
5. Hyperbolic Paraboloid	$\frac{x^2}{a^2}-\frac{y^2}{b^2}=2cz$	Saddle-shaped non-central conicoid; doubly ruled surface.

Problem 24.1 (PUST Past Exam): Reduce $2x^2-7y^2+2z^2-10yz-8zx-10xy+6x+12y-6z+2=0$ to standard form.

Solution (Exam Style):

$$\text{Discriminating cubic: } \begin{vmatrix} 2-\lambda & -5 & -4 \\ -5 & -7-\lambda & -5 \\ -4 & -5 & 2-\lambda \end{vmatrix} = 0$$

$$\implies \lambda^3+3\lambda^2-54\lambda-216=0$$

$$\implies (\lambda-6)(\lambda+3)(\lambda+12)=0 \implies \lambda=6,\ -3,\ -12.$$

Solving $\frac{1}{2}\frac{\partial F}{\partial x}=0$, $\frac{1}{2}\frac{\partial F}{\partial y}=0$, $\frac{1}{2}\frac{\partial F}{\partial z}=0$:

$$(\alpha,\beta,\gamma)=\left(\frac{1}{3},\frac{1}{3},\frac{1}{2}\right), \qquad d'=u\alpha+v\beta+w\gamma+d=\frac{7}{2}$$

$$\therefore 6X^2-3Y^2-12Z^2+\frac{7}{2}=0 \quad \text{[Hyperboloid of One Sheet]}.$$

Quick Revision — Module 8 (Units 22–24)

- **Sphere:** Center $(-u,-v,-w)$, $R=\sqrt{u^2+v^2+w^2-d}$; Circle section radius $r=\sqrt{R^2-p^2}$; Family through circle: $S+\lambda P=0$.
- **Right Circular Cone:** $[l(x-\alpha)+m(y-\beta)+n(z-\gamma)]^2=(l^2+m^2+n^2)[(x-\alpha)^2+(y-\beta)^2+(z-\gamma)^2]\cos^2\theta$.
- **Right Circular Cylinder:** $(x-\alpha)^2+(y-\beta)^2+(z-\gamma)^2-\frac{[l(x-\alpha)+m(y-\beta)+n(z-\gamma)]^2}{l^2+m^2+n^2}=R^2$.

Common Mistakes to Avoid in the Exam

- Confusing the sphere radius R with the plane-section circle radius $r=\sqrt{R^2-p^2}$, or forgetting to normalize the axis DRs (l,m,n) when computing projection length on a cylinder's axis.